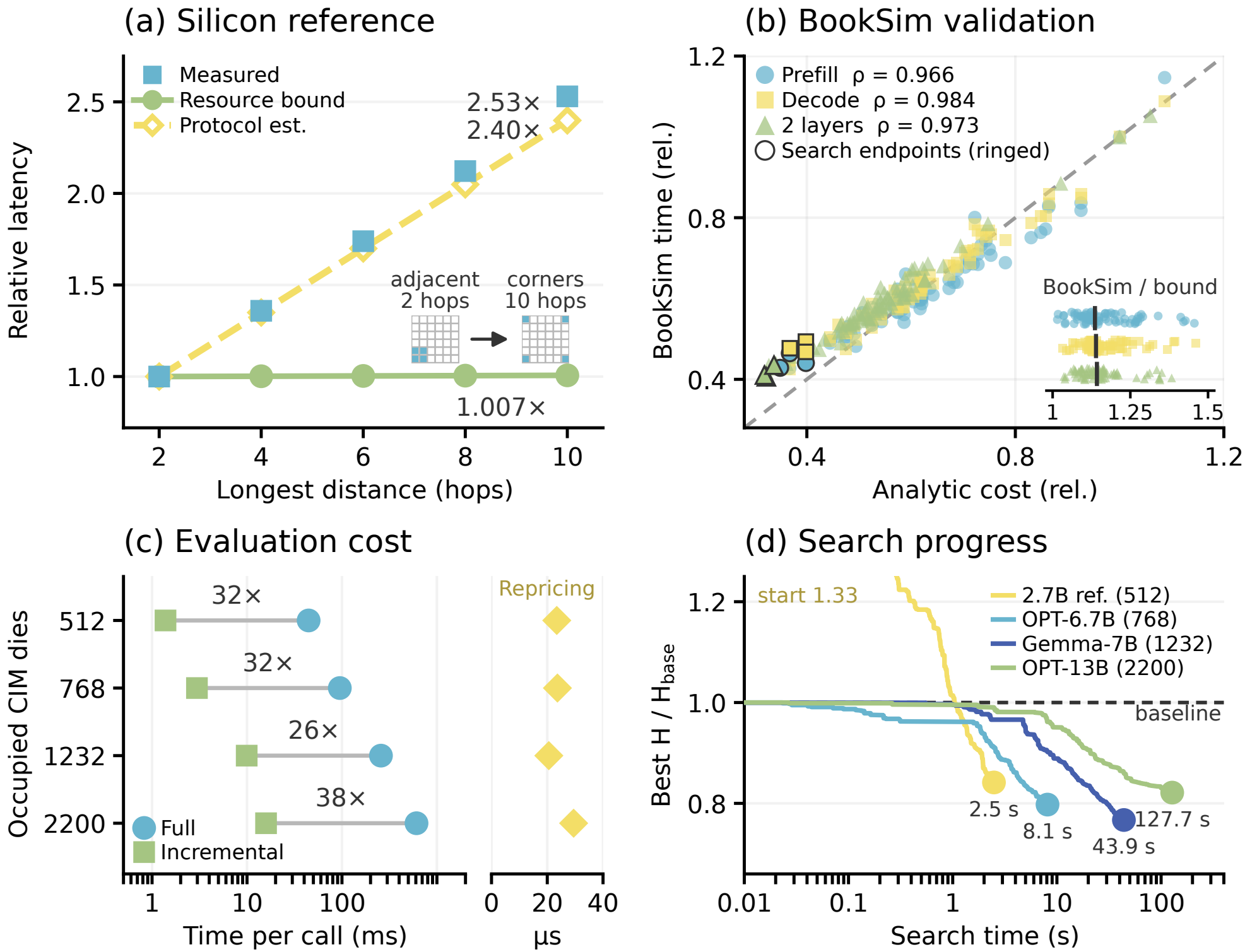
(a) Silicon reference
Measured
Resource bound
Protocol est.
2.53×
2.40×
1.007×
adjacent
2 hops
corners
10 hops
Relative latency
Longest distance (hops)
(b) BookSim validation
Prefill ρ = 0.966
Decode ρ = 0.984
2 layers ρ = 0.973
Search endpoints (ringed)
BookSim / bound
BookSim time (rel.)
Analytic cost (rel.)
(c) Evaluation cost
32×
32×
26×
38×
Full
Incremental
Repricing
Occupied CIM dies
Time per call (ms)
µs
(d) Search progress
start 1.33
2.7B ref. (512)
OPT-6.7B (768)
Gemma-7B (1232)
OPT-13B (2200)
baseline
2.5 s
8.1 s
43.9 s
127.7 s
Best H / H_base
Search time (s)